

Flux-Tunable Cavity for Dark Matter Detection

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Developing a dark matter detector with wide mass tunability is an immensely desirable property, yet, it is challenging due to maintaining strong sensitivity. Resonant cavities for dark matter detection have traditionally employed mechanical tuning, moving parts around to change electromagnetic boundary conditions. However, these cavities have proven challenging to operate in sub-Kelvin cryogenic environments due to differential thermal contraction, low heat capacities, and low thermal conductivities. Instead, we develop an electronically tunable cavity architecture by coupling a superconducting 3D microwave cavity with a dc flux tunable superconducting quantum interference device. With a flux delivery system engineered to maintain high coherence in the cavity, we perform a hidden-photon dark matter search below the quantum-limited threshold. A microwave photon counting technique is employed through repeated quantum nondemolition measurements using a transmon qubit. With this device, we perform a hidden-photon search and constrain the kinetic mixing angle to $\epsilon < 8.2 \times 10^{-15}$ in a tunable band from 5.672 to 5.694 GHz. By coupling multimode tunable cavities to the transmon, wider hidden-photon searching ranges are possible.

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Introduction—Dark matter (DM) is an integral part of the standard model of cosmology, though its specific composition is still a mystery. If it is comprised of particles with mass less than ~ 100 eV, the inferred galactic DM mass density implies that the mode occupation number must be large [1]. In this mass range, DM may be modeled as classical, coherent waves that can exert weak, sinusoidal forces on laboratory oscillators with signal frequency f . Since dark matter is nonrelativistic, the signal frequency is proportional to its rest mass m [2]. The strength of these forces is unknown; while it must be weaker than electromagnetism because the DM is not luminous, since dark matter has only been observed via its gravitational imprint on cosmology and astrophysics, the couplings to mass energy could be as weak as gravity [3] and still reproduce all cosmological observations. A large range of allowed

coupling strengths remains unexplored between these two limits.

Direct detection experiments perform radio searches for wavelike DM by tuning a laboratory oscillator over a range of resonant frequencies to search for excess noise power from DM disturbances above that estimated from data in neighboring frequency tunings [4]. The detection bandwidth $b = 10^{-6}f$ is determined by the kinetic energy spread of the distribution of DM trapped well in the gravitational potential of the galaxy with maximum velocity limited by the escape velocity [5]. In the small signal limit, individual signal photons in the oscillator can be detected using single-photon counters. Because these counters are insensitive to the phase of the signal oscillation, they can evade the standard quantum limit and reach noise levels limited only by Poisson fluctuations of signal and background counts.

With standard assumptions about the local DM density ρ [5], a signal excess or absence at a particular frequency tuning can be interpreted as a detection or constraint on the DM coupling to photons. For DM in the form of dark or hidden photons (HPs) [6,7], the driving force can be modeled as an effective current, which is given by

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$$j_{\text{HP}} = \varepsilon m_{\gamma'} \sqrt{2\rho} e^{im_{\gamma'} \hat{u}}. \quad (1)$$

Here, ε is the postulated kinetic mixing angle between standard electromagnetism and hidden sector electromagnetism, which determines the strength of the coupling, and the hidden electric field amplitude of the DM wave is $E' = \sqrt{2\rho}$. Here, the polarization of the hidden-photon field is denoted by $\hat{u}$, and the mass of the hidden-photon is represented by $m_{\gamma'}$. A microwave cavity with a resonance frequency f tuned to the hidden-photon mass $m_{\gamma'} = hf$ is used to coherently accumulate the electromagnetic response to the extremely undercoupled DM driving force. Here, h is the Planck constant. Because the DM mass is unknown, it is critical to build detectors with wide frequency tunability.

Quantum techniques have brought unprecedented sensitivities in single microwave photon detection for DM searches [8–13]. Recently, the repeated quantum nondemolition measurements of fixed-frequency cavity photons through a superconducting qubit achieved a dark count rate of around $1/s$ [8]. The qubit-cavity interaction can also prepare non-Gaussian initial quantum states, which enhance the signal photon rate via stimulated emission [9]. Balembois *et al.* present an itinerant single microwave photon receiver based on an irreversible four-wave mixing process with a dark count rate 85 s^{-1} and 50 MHz tunability [10,11]. DM searches are also possible by looking for the forces exerted directly on a qubit whose frequency is tunable using a superconducting quantum interference device (SQUID) [12,13]. However, the conductive rf-shielding enclosure may strongly suppress the electromagnetic response with a spectrum of modes at fixed frequencies.

Here, we use a dc flux-tunable SQUID to directly tune the frequency of a coupled cryogenic 3D microwave cavity. The flux delivery system is designed to maintain the cavity coherence required to accumulate and store severely undercoupled DM signals and to reject out-of-band noise. Unlike the tunable 2D resonators and tunable qubits concepts mentioned above, the 3D storage cavity offers a significantly larger volume, making it an ideal target for intercepting dark matter waves. Theoretically, the 3D cavity mode frequency can have a tuning range as large as the frequency separation in the avoided level crossing [14] between the storage and the coupler modes, approximately $2g \sim 200 \text{ MHz}$, where g is the coupling strength between storage and coupler modes. Operating multiple tunable cavities simultaneously expands the overall tuning range for DM experiments.

The structure of the Letter is organized as follows: First, we present the theoretical framework and describe the experimental setup of the tunable photon counting detector. Second, we perform an experimental calibration of the efficiency of the detector within the detection band. Finally, we analyze the results of the hidden-photon

exclusion experiments and propose directions for future improvements.

Tunable photon counting detector—Our tunable single-photon counter comprises four main components [see Fig. 1(a)]: a storage cavity; a coupler chip that includes both an on-chip flux filter and a dc SQUID; a qubit chip; and a readout cavity, which shares the same “Coffin” cavity as the storage. Each component hosts a single mode utilized in the experiment: A high- Q ($Q > 10^6$) tunable storage mode ω_s that stores single-photon signals in the cavity [15], a symmetric dc SQUID mode ω_c that tunes the storage mode frequency while maintaining cavity coherence, a dispersively coupled transmon ω_q [16], and a readout mode ω_r which implements the quantum nondemolition (QND) measurement of the storage photons [17,18]. The Hamiltonian of the system is

$$\begin{aligned} H(\Phi_{\text{ext}}) = & \omega_s(\Phi_{\text{ext}}) a_s^\dagger a_s + \omega_q a_q^\dagger a_q + \omega_c(\Phi_{\text{ext}}) a_c^\dagger a_c + \omega_r a_r^\dagger a_r \\ & + \frac{\alpha_q}{2} a_q^\dagger a_q^\dagger a_q a_q + 2\chi(\Phi_{\text{ext}}) n_q n_s \\ & + 2\chi_{qr} n_q n_r + 2\chi_{sc}(\Phi_{\text{ext}}) n_s n_c, \end{aligned} \quad (2)$$

Here, a and n are the annihilation and photon number operators with s, q, c , and r denoting the storage, transmon, coupler, and readout modes, respectively. The transmon-storage, transmon-readout, and storage-coupler coupling strengths are denoted by $\{\chi, \chi_{qr}, \chi_{sc}\}$, respectively, and Φ_{ext} is the dc flux through the coupler SQUID loop. For simplicity, Φ_{ext} is expressed in units of the magnetic flux quantum Φ_0 . Figure 1(b) shows the storage mode photon number resolved peaks with $\chi = -2\pi \times 1.655 \text{ MHz}$ when $\Phi_{\text{ext}} = 0$. The photon-resolved χ is necessary to map storage mode signals to the transmon state efficiently, then dispersively read out through the transmon-readout coupling $\chi_{qr} = -2\pi \times 0.14 \text{ MHz}$. Here, the linewidths $\{\kappa_q, \kappa_s\}$ for the qubit and storage are separately $\{4.7, 2.4\} \text{ kHz}$. The dc flux tunable coupler mode ω_c is capacitively coupled to the storage mode, which allows for tuning ω_s by biasing ω_c . The theoretical tuning range for our detector can reach up to twice the strength of the capacitive coupling, which is on the order of 100 MHz (See Supplemental Material II for details [19]). The other stray couplings (coupler transmon, coupler readout) are negligible; Therefore, the transmon frequency, readout frequency, and transmon anharmonicity $\{\omega_q, \omega_r, \alpha_q\} = 2\pi \times \{4811, 6773, -176\} \text{ MHz}$ are approximated as constants in theory. Figure 1(c) displays a comparison between black-box quantization [23] and experimental spectroscopy data, showing a good match across the dc flux region ($\Phi_{\text{ext}} \in [-0.4793\pi, 0]$) where our scheme is implemented. This corresponds to a detector tuning range of 22.8 MHz in our experiment. The SQUID frequency limits the tuning range: in our case, the SQUID frequency is too high to see the full storage coupler avoided

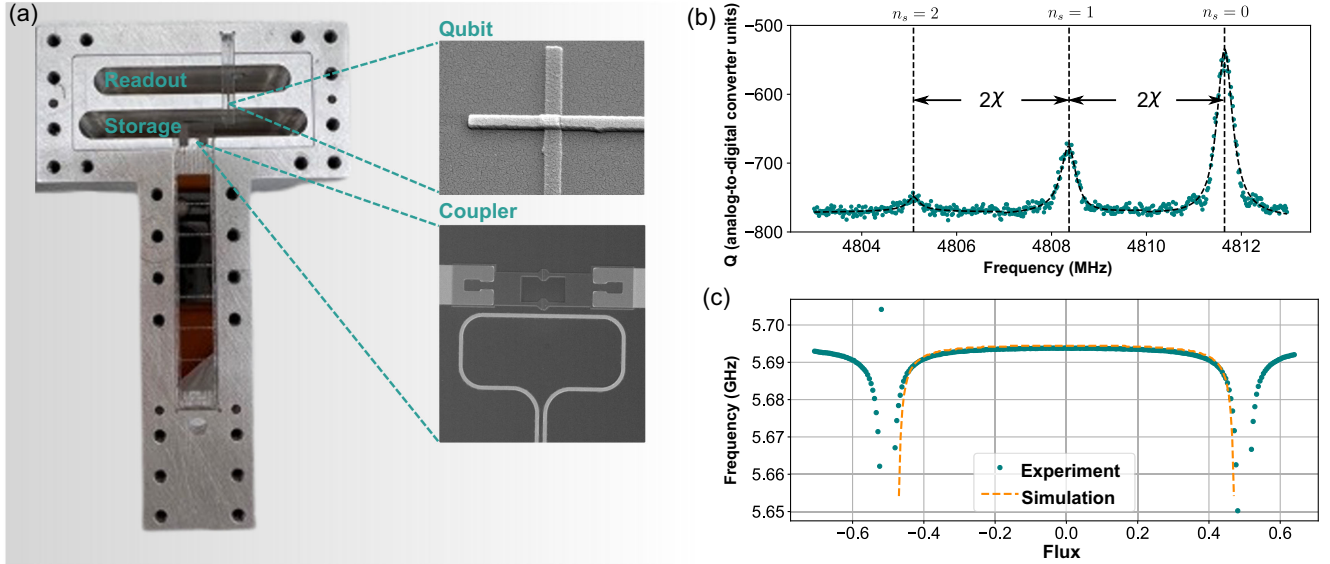


FIG. 1. Tunable superconducting qubit hidden-photon detector. (a) The tunable hidden-photon detector comprises a Coffin cavity with two modes: readout (top) and storage (bottom). The qubit chip (top) is capacitively coupled to both modes, and the coupler chip (bottom) is coupled only to the storage mode for frequency tuning. The insets show the scanning electron micrograph of the qubit junction (top) and the SQUID loop (bottom). (b) Qubit frequency-resolved two-tone spectroscopy for a displaced state in the storage mode at $\Phi_{\text{ext}} = 0$. In this measurement, the qubit is weakly driven while monitoring the readout; a response shift indicates the drive matches the qubit frequency modified by storage photons. Dashed lines show the corresponding peak positions when the storage is at $\{|0\rangle, |1\rangle, |2\rangle\}$, respectively. (c) dc flux scan of storage mode single-tone spectroscopy. The storage mode peak positions are extracted and shown in the green dots. The black-box simulation [23] results are included in the yellow dashed line.

crossing within a dc flux range that maintains a reasonable storage dephasing time.

Figure 2(a) shows our photon counting scheme using the Ramsey interferometry readout [8], which detects the parity of the storage mode photon population. For each new detection frequency, ω_s is dc flux biased according to the flux-frequency mapping shown in Fig. 1(c). For flux points requiring detector efficiency calibration, we initiate the experiment by adding photons to the storage cavity. For actual hidden-photon detection, the experiments begin with a waiting period corresponding to the signal integration time, instead. Following this procedure, N repeated parity measurements are performed: In each parity measurement, a readout tone (2.3 μs long) is first sent to check the current transmon state in $|g\rangle$ or $|e\rangle$. This is followed by a 5 μs pause to allow for the readout to reset. Subsequently, a fast $\pi/2$ transmon rotation is applied using a 2σ tailed Gaussian pulse [where $\sigma = 10 \text{ ns} \ll 1/\chi(\Phi_{\text{ext}})$], positioning the transmon into a superposition state $(|g\rangle \pm |e\rangle)/\sqrt{2}$. The transmon then waits for $t_p = \pi/|2\chi|$ to acquire a phase that depends on the storage state. After another $\pi/2$ rotation, the transmon state is flipped if the storage mode is in an odd-number Fock state. A positive detection of a photon is indicated by observing a repeated sequence of transmon state transitions during the N parity measurements. In our detection protocol, we assume that the signals received by the storage mode are sufficiently weak, focusing only on the population within the first two levels of the storage

mode. Therefore, the transmon state is flipped when the storage mode is excited.

To resolve all possible error mechanisms during the measurement protocol, we model the evolution of the storage cavity, transmon, and readout as a hidden Markov process where the cavity and transmon states are hidden variables that emit as a readout signal [8], and the storage initial state probability $\{P_0, P_1\}$ is reconstructed using the backward algorithm. Here, P_0 and $P_1 = 1 - P_0$ represent the calculated probabilities of the storage mode being in the states $|0\rangle$ and $|1\rangle$. In our experiment, we define a state discrimination threshold $\lambda_{\text{thresh}} = P_1/P_0$, and the false positive probability δ is exponentially suppressed as λ_{thresh} increases. The storage mode is considered in $|1\rangle$ when $\lambda_{\text{thresh}} \geq 125$. Details about the hidden Markov process analysis and choosing λ_{thresh} are discussed in Supplemental Material IV [19].

The storage and readout modes are machined on the same high-purity (6N) Aluminum cavity. Details about the cavity etching are discussed in Supplemental Material I [19]. An edge-coupled microstrip on-chip filter is used on the flux-biasing line to protect the storage mode coherence. Details about the on-chip flux filter design are shown in Supplemental Material II [19].

Detector efficiency characterization—The tunable photon counting detector efficiency η is characterized by initially injecting different predetermined populations of photons $\bar{n}_{\text{inj}}$ into the storage mode through weak displacement drives. The weak storage displacement β ($\beta \ll 1$) ensures the

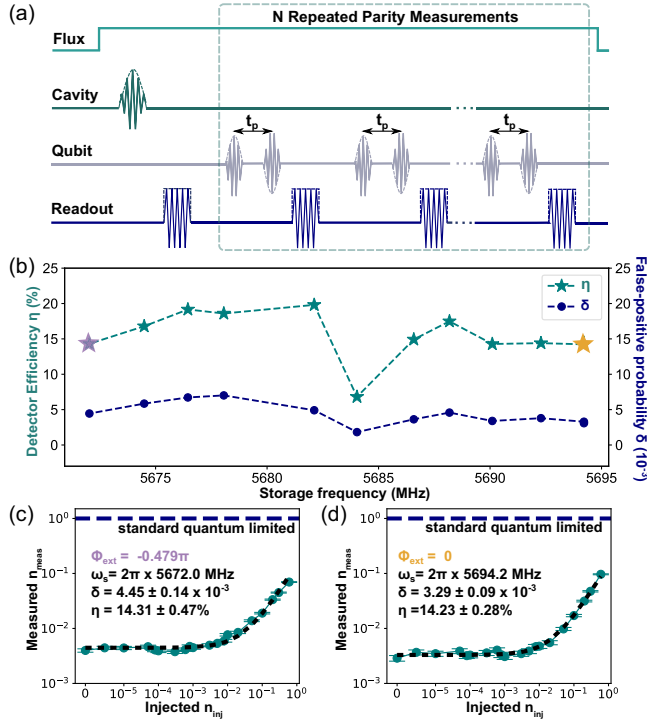


FIG. 2. (a) Photon counting scheme. The pulse sequence at each dc flux bias point includes an initial weak storage displacement for detector efficiency characterization or a hidden-photon signal integration period. N repeated parity measurements are performed afterward, containing a readout pulse, readout photon releasing period, and two qubit $\pi/2$ rotations gapped by idling period $t_p = \pi/|2\chi|$. N bits of single shot readout after each detection cycle are collected. The probability of seeing false-positive events from the detector is exponentially suppressed in our detection scheme. (b) Detector efficiency characterization. At each flux biasing point, a list of precalibrated weak displacement drives is separately applied to the storage, followed by $N = 25$ parity measurements and Markovian analysis with threshold $\lambda_{\text{thresh}} = 125$. η and δ are determined from the linear fitting. At $\Phi_{\text{ext}} = 0$, the thermal population of the storage is quantified using the fast Fourier transformation (FFT) of the transmon Ramsey signal. Two detector characterizations ($\Phi_{\text{ext}} = -0.479\pi$ and 0) are shown in (c). More characterization data are shown in Supplemental Material IV [19].

population remains mostly in the Fock states $\{|0\rangle, |1\rangle\}$. The calibration of the storage mode displacement is discussed in Supplemental Material III [19]. Then, the detector efficiency can be determined through the proportion of positive events $\bar{n}_{\text{meas}}$ after the Ramsey interferometry readout: We choose $N = 25$ to discriminate between one and zero photon events in the storage mode. By varying $\bar{n}_{\text{inj}}$, we fit this relationship with the function $\bar{n}_{\text{meas}} = \eta \bar{n}_{\text{inj}} + \delta$ to extract the detector efficiency η and false positive probability δ , defined as the background excitations in the storage detected by our photon counting scheme.

Figure 2(b) displays the detector characterization results across a range of flux tuning points. Two detailed detector

efficiency plots at two flux tuning points are shown in Fig. 2(c). The other characterizations of the flux points are in Supplemental Material IV [19]. The detector efficiency remains reliably above 14%, even when Φ_{ext} is close to $\pi/2$. This consistent performance is attributed to three key factors: First, the transmon and storage T_1 are insensitive to Φ_{ext} , which are protected by the on-chip flux line filter as detailed in Supplemental Material II [19]. Second, the tunable hidden-photon detector is resilient against transmon bit-flip error through continuous parity measurements. The hidden-photon Markov process analysis (see below) also compensates for variations in the storage decay time T_1^s between tuning points.

The primary limitation in detector performance arises from storage cavity coherence and the background of real photons. We perform the master equation simulation to capture the detector performance (See Supplemental Material V [19]) and show that the detector efficiency is limited to 25% with the coherence of our current setup. The storage coherence time limits N before the stored photons decay. Spurious excitations of photons in the storage mode alter the parity readout pattern, leading to false positive events. δ increases as Φ_{ext} is tuned closer to $\pi/2$. We suspect that the enhanced hybridization between the coupler and storage modes near $\pi/2$ accelerates the transfer of the quasiparticle tunneling events of the coupler into storage excitations. Additionally, the thermal population of the storage at $\Phi_{\text{ext}} = 0$ is quantified using the FFT of the transmon Ramsey signal. The measured intensity ratio at $\Phi_{\text{ext}} = 0$ for transmon frequency at ω_q and $\omega_q - \chi$ is $8.7 \pm 0.5 \times 10^{-3}$ [24]. This measurement suggests that the storage thermal population $\bar{n}_c = 8.6 \pm 0.5 \times 10^{-3}$, corresponding to a cavity temperature of 57 ± 1.4 mK. We choose the detection threshold $\lambda_{\text{thresh}} = 125$ for varying flux tuning points [8] such that the background photon probability $[1/(\lambda_{\text{thresh}} + 1) < \bar{n}_c]$ is comparable to the storage thermal population.

The measured storage temperature exceeds the mixing chamber temperature (15 mK), possibly because of quasiparticle tunneling events in the SQUID and the transmon being projected into stored photons via the dressing of the cavity [8]. Gap engineering [25], quasiparticle trapping [26], and comprehensive infrared radiation shielding [27] can effectively mitigate these effects.

Hidden-photon dark matter exclusion—By measuring photon counts through repeated parity assessments and utilizing a Markov model for analysis, we conduct a hidden-photon dark matter search over a 22.8 MHz range. We sample the tuning range with 12 695 dc flux tuning points, achieving a frequency resolution of 2.8 kHz for 83.7% of the tuning steps, commensurate with half the dark matter linewidth (5.7 kHz) [28]. Smaller step sizes could be achieved in future experiments by increasing the attenuation on the dc line inside the fridge. We collect 20 000 independent parity measurements at each tuning point. For each measurement, we perform $N = 25$ Ramsey interferometry sequences with a readout length $t_r = 2.3$ μ s,

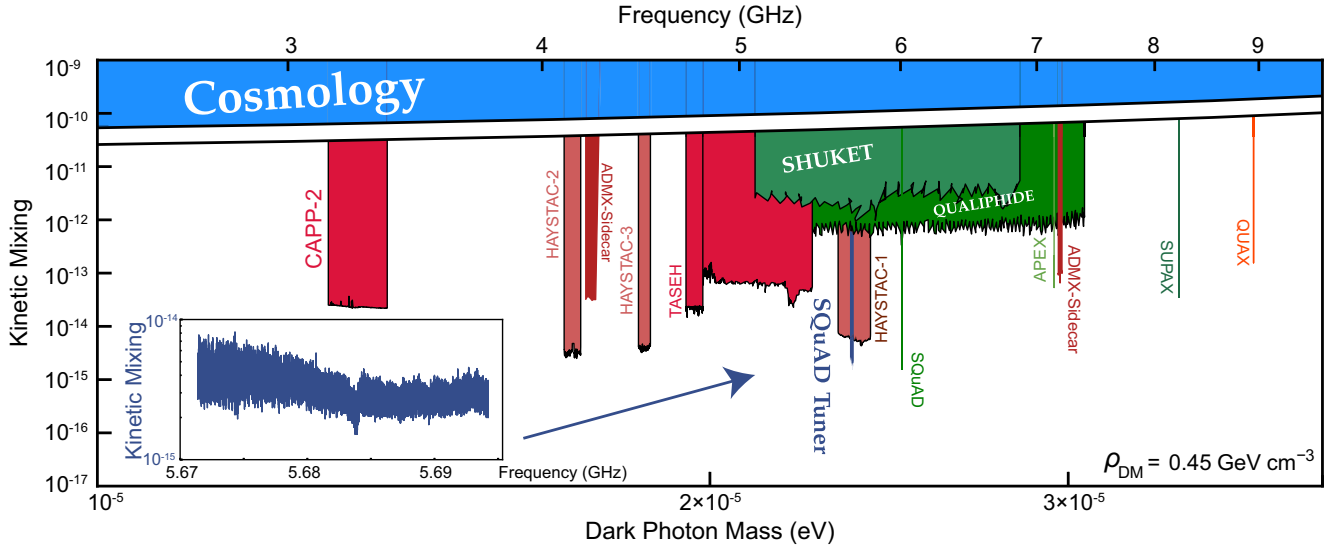


FIG. 3. Comparison with previous dark photon exclusion results [8,30–44]. Our exclusion data is labeled as “SQuAD_Tuner.” Inset: hidden-photon exclusion results measured in this experiment. The storage cavity frequency is swept by 22.8 MHz, and at each frequency point, 20 000 independent parity measurements are collected. The exclusion results at 95% confidence level are analyzed based on Reference [8]. All flux-dependent parameters are linearly interpolated and used in the analysis. The plot is generated using the code from Ref. [45]. The geometric factor is not included in the plot for comparison.

readout photon releasing time $t_l = 5 \mu\text{s}$, system reset time and signal integration time $t_c = 500 \mu\text{s}$. For detector characterization, the values of t_c are chosen to be greater than five times the T_1 of the storage mode in our setup. Within the same dataset, for the data with the calibration signal source turned off, the dark matter storage signal integration time t_i is assumed to be only the storage lifetime $T_1^s \in [64.5, 69.2] \mu\text{s}$. Thus, each measurement cycle time is $t_c + (t_r + t_l) \times 25 = 682.5 \mu\text{s}$ with $182.5 \mu\text{s}$ used for the readout sequence. The system reset time is required during the calibration process; however, in the actual detection experiment, no reset is necessary, and the entire duration t_c is dedicated to signal integration. The experiment has a duty cycle of 27%, with a measurement time of 10.65 s per flux tuning point. The 22.8 MHz scan is completed in 48.1 hours, enabling future GHz-range tuning to finish within a reasonable amount of time.

We choose a detector threshold of $\lambda_{\text{thresh}} = 125$ and count a maximum of 91 photons at 5680.2859 MHz as seen in Fig. 3. To search for hidden-photon candidates in the tuning range [5671.395, 5694.212] MHz, first, we obtain a photon count background by smoothing the counts as a function of frequency using a Savitzky-Golay filter of order four and window length of 112 [29]. We expect a dark matter signal to be distributed over two frequency bins and ensure the filter window size is much larger than the potential signal. We compute the cumulative probability of a dark matter candidate producing the observed signal above the measured background. This cumulative probability calculation takes the form $P(\leq N_{\text{obs}}) = 1 - \sum_{N_{\text{DM}}=0}^{N_{\text{obs}}} [(N_{\text{back}} + N_{\text{DM}})^{N_{\text{obs}}} e^{-(N_{\text{back}} + N_{\text{DM}})} / N_{\text{obs}}!]$. Additionally, we account for the systematic uncertainties by

marginalizing over the relevant measured parameters (See Supplemental Material VI [19] for details). Dark matter candidates that could produce a signal less than or equal to the observed counts with less than 5% probability are excluded at the 95% confidence level. See Supplemental Material [19] for details on determining and accounting for the background level and incorporating systematics into the 95% confidence level exclusion of dark matter hypothesis. Figure 3 shows hidden-photon candidates that we exclude at the 95% confidence level in the context of other experiments sensitive to wavelike dark matter.

Conclusion—In this Letter, we demonstrate a state-of-the-art electronically tunable photon counter for dark matter detection, achieving an approximately 20-fold speedup compared to a linear quantum-limited amplifier [8,46]. By coupling to a dc-tunable SQUID, we realized a tunability of 22.8 MHz in our storage cavity. Future devices should be able to use the level repulsion [14] at the avoided crossing to achieve the entire tuning range of ~ 200 MHz; the demonstration of SQUID-based tuning of a 3D cavity provides a promising avenue for future dark matter searches with GHz tuning range, employing many such tunable cavity modes, each with 3% tuning range. The instantaneous tuning range can also be expanded by coupling a single SQUID to a multimode cavity system. This enables concurrent tuning of each cavity mode to search for the dark matter at each mode frequency simultaneously.

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Data availability—The data that support the findings of this article are openly available [47], embargo periods may apply.

- [1] L. Randall, J. Scholtz, and J. Unwin, Cores in dwarf galaxies from Fermi repulsion, *Mon. Not. R. Astron. Soc.* **467**, stx161 (2017).
- [2] P. Sikivie, Experimental tests of the “invisible” axion, *Phys. Rev. Lett.* **51**, 1415 (1983).
- [3] D. Carney, S. Ghosh, G. Krnjaic, and J. M. Taylor, Proposal for gravitational direct detection of dark matter, *Phys. Rev. D* **102**, 072003 (2020).
- [4] P. W. Graham, I. G. Irastorza, S. K. Lamoreaux, A. Lindner, and K. A. van Bibber, Experimental searches for the axion and axion-like particles, *Annu. Rev. Nucl. Part. Sci.* **65**, 485 (2015).
- [5] D. Baxter *et al.*, Recommended conventions for reporting results from direct dark matter searches, *Eur. Phys. J. C* **81**, 907 (2021).
- [6] P. W. Graham, J. Mardon, and S. Rajendran, Vector dark matter from inflationary fluctuations, *Phys. Rev. D* **93**, 103520 (2016).
- [7] S. Chaudhuri, P. W. Graham, K. Irwin, J. Mardon, S. Rajendran, and Y. Zhao, Radio for hidden-photon dark matter detection, *Phys. Rev. D* **92**, 075012 (2015).
- [8] A. V. Dixit, S. Chakram, K. He, A. Agrawal, R. K. Naik, D. I. Schuster, and A. Chou, Searching for dark matter with a superconducting qubit, *Phys. Rev. Lett.* **126**, 141302 (2021).
- [9] A. Agrawal, A. V. Dixit, T. Roy, S. Chakram, K. He, R. K. Naik, D. I. Schuster, and A. Chou, Stimulated emission of signal photons from dark matter waves, *Phys. Rev. Lett.* **132**, 140801 (2024).
- [10] L. Balembois, J. Travesedo, L. Pallegoix, A. May, E. Billaud, M. Villiers, D. Estève, D. Vion, P. Bertet, and E. Flurin, Cyclically operated microwave single-photon counter with sensitivity of 10^{-22} W/ $\sqrt{\text{Hz}}$, *Phys. Rev. Appl.* **21**, 014043 (2024).
- [11] C. Braggio, L. Balembois, R. D. Vora, Z. Wang, J. Travesedo, L. Pallegoix, G. Carugno, A. Ortolan, G. Ruoso, U. Gambardella, D. D’Agostino, P. Bertet, and E. Flurin, Quantum-enhanced sensing of axion dark matter with a transmon-based single microwave photon counter, *arXiv:2403.02321*.
- [12] S. Chen, H. Fukuda, T. Inada, T. Moroi, T. Nitta, and T. Sihanugrist, Detecting hidden photon dark matter using the direct excitation of transmon qubits, *Phys. Rev. Lett.* **131**, 211001 (2023).
- [13] S. Chen, H. Fukuda, T. Inada, T. Moroi, T. Nitta, and T. Sihanugrist, Search for QCD axion dark matter with transmon qubits and quantum circuit, *Phys. Rev. D* **110**, 115021 (2024).
- [14] J. Majer, J. M. Chow, J. M. Gambetta, J. Koch, B. R. Johnson, J. A. Schreier, L. Frunzio, D. I. Schuster, A. A. Houck, A. Wallraff, A. Blais, M. H. Devoret, S. M. Girvin, and R. J. Schoelkopf, Coupling superconducting qubits via a cavity bus, *Nature (London)* **449**, 443 (2007).
- [15] R. Barbieri, C. Braggio, G. Carugno, C. Gallo, A. Lombardi, A. Ortolan, R. Pengo, G. Ruoso, and C. Speake, Searching for galactic axions through magnetized media: The QUAX proposal, *Phys. Dark Universe* **15**, 135 (2017).
- [16] J. Koch, T. M. Yu, J. Gambetta, A. A. Houck, D. I. Schuster, J. Majer, A. Blais, M. H. Devoret, S. M. Girvin, and R. J. Schoelkopf, Charge-insensitive qubit design derived from the Cooper pair box, *Phys. Rev. A* **76**, 042319 (2007).
- [17] M. Brune, S. Haroche, V. Lefevre, J. M. Raimond, and N. Zagury, Quantum nondemolition measurement of small photon numbers by Rydberg-atom phase-sensitive detection, *Phys. Rev. Lett.* **65**, 976 (1990).
- [18] S. Gleyzes, S. Kuhr, C. Guerlin, J. Bernu, S. Deléglise, U. Busk Hoff, M. Brune, J.-M. Raimond, and S. Haroche, Quantum jumps of light recording the birth and death of a photon in a cavity, *Nature (London)* **446**, 297 (2007).
- [19] See Supplemental Material at <http://link.aps.org/supplemental/10.1103/clp9-xc2n> for additional experimental details and derivations, which include Refs. [20–22].
- [20] S. Chaudhuri, P. W. Graham, K. Irwin, J. Mardon, S. Rajendran, and Y. Zhao, Radio for hidden-photon dark matter detection, *Phys. Rev. D* **92**, 075012 (2015).
- [21] A. Read, Modified frequentist analysis of search results (the CL_s method), Report No. CERN-OPEN-2000-205, CERN, 2000.
- [22] J. W. Foster, N. L. Rodd, and B. R. Safdi, Revealing the dark matter halo with axion direct detection, *Phys. Rev. D* **97**, 123006 (2018).
- [23] S. E. Nigg, H. Paik, B. Vlastakis, G. Kirchmair, S. Shankar, L. Frunzio, M. H. Devoret, R. J. Schoelkopf, and S. M. Girvin, Black-box superconducting circuit quantization, *Phys. Rev. Lett.* **108**, 240502 (2012).
- [24] A. P. Sears, A. Petrenko, G. Catelani, L. Sun, H. Paik, G. Kirchmair, L. Frunzio, L. I. Glazman, S. M. Girvin, and R. J. Schoelkopf, Photon shot noise dephasing in the strong-dispersive limit of circuit QED, *Phys. Rev. B* **86**, 180504 (2012).
- [25] L. Sun, L. DiCarlo, M. D. Reed, G. Catelani, L. S. Bishop, D. I. Schuster, B. R. Johnson, G. A. Yang, L. Frunzio, L. Glazman, M. H. Devoret, and R. J. Schoelkopf, Measurements of quasiparticle tunneling dynamics in a band-gap-engineered transmon qubit, *Phys. Rev. Lett.* **108**, 230509 (2012).
- [26] X. Pan, Y. Zhou, H. Yuan, L. Nie, W. Wei, L. Zhang, J. Li, S. Liu, Z. H. Jiang, G. Catelani, L. Hu, F. Yan, and D. Yu, Engineering superconducting qubits to reduce quasiparticles and charge noise, *Nat. Commun.* **13**, 7196 (2022).
- [27] J. M. Kreikebaum, A. Dove, W. Livingston, E. Kim, and I. Siddiqi, Optimization of infrared and magnetic shielding of superconducting TiN and Al coplanar microwave resonators, *Supercond. Sci. Technol.* **29**, 104002 (2016).
- [28] M. S. Turner, Periodic signatures for the detection of cosmic axions, *Phys. Rev. D* **42**, 3572 (1990).

- [29] A. Yi, S. Ahn, B. Ko, and Y. K. Semertzidis, Analytical estimation of the signal to noise ratio efficiency in axion dark matter searches using a Savitzky-Golay filter, *J. High Energy Phys.* **11** (2023) 115.
- [30] S. J. Witte, S. Rosauero-Alcaraz, S. D. McDermott, and V. Poulin, Dark photon dark matter in the presence of inhomogeneous structure, *J. High Energy Phys.* **06** (2020) 132.
- [31] A. Caputo, H. Liu, S. Mishra-Sharma, and J. T. Ruderman, Dark photon oscillations in our inhomogeneous universe, *Phys. Rev. Lett.* **125**, 221303 (2020).
- [32] P. Arias, D. Cadamuro, M. Goodsell, J. Jaeckel, J. Redondo, and A. Ringwald, Wispy cold dark matter, *J. Cosmol. Astropart. Phys.* **06** (2012) 013.
- [33] P. Brun, L. Chevalier, and C. Flouzat, Direct searches for hidden-photon dark matter with the SHUKET experiment, *Phys. Rev. Lett.* **122**, 201801 (2019).
- [34] K. Ramanathan, N. Klimovich, R. Basu Thakur, B. H. Eom, H. G. Leduc, S. Shu, A. D. Beyer, and P. K. Day, Wideband direct detection constraints on hidden photon dark matter with the qualiphide experiment, *Phys. Rev. Lett.* **130**, 231001 (2023).
- [35] C. Boutan *et al.* (ADMX Collaboration), Piezoelectrically tuned multimode cavity search for axion dark matter, *Phys. Rev. Lett.* **121**, 261302 (2018).
- [36] L. Zhong *et al.*, Results from phase 1 of the HAYSTAC microwave cavity axion experiment, *Phys. Rev. D* **97**, 092001 (2018).
- [37] K. M. Backes *et al.*, A quantum enhanced search for dark matter axions, *Nature (London)* **590**, 238 (2021).
- [38] M. J. Jewell *et al.* (HAYSTAC Collaboration), New results from HAYSTAC's phase II operation with a squeezed state receiver, *Phys. Rev. D* **107**, 072007 (2023).
- [39] J. Jeong, S. Youn, S. Bae, J. Kim, T. Seong, J. E. Kim, and Y. K. Semertzidis, Search for invisible axion dark matter with a multiple-cell haloscope, *Phys. Rev. Lett.* **125**, 221302 (2020).
- [40] C. M. Adair *et al.*, Search for dark matter axions with CAST-CAPP, *Nat. Commun.* **13**, 6180 (2022).
- [41] H. Chang *et al.* (TASEH Collaboration), First results from the Taiwan axion search experiment with a haloscope at 19.6 μeV , *Phys. Rev. Lett.* **129**, 111802 (2022).
- [42] D. Alesini *et al.*, Galactic axions search with a superconducting resonant cavity, *Phys. Rev. D* **99**, 101101 (2019).
- [43] T. Schneemann, K. Schmieden, and M. Schott, First results of the SUPAX experiment: Probing dark photons, [arXiv:2308.08337](https://arxiv.org/abs/2308.08337).
- [44] S. Abrahamyan *et al.*, Search for a new gauge boson in electron-nucleus fixed-target scattering by the APEX experiment, *Phys. Rev. Lett.* **107**, 191804 (2011).
- [45] A. Caputo, A. J. Millar, C. A. J. O'Hare, and E. Vitagliano, Dark photon limits: A handbook, *Phys. Rev. D* **104**, 095029 (2021).
- [46] S. K. Lamoreaux, K. A. van Bibber, K. W. Lehnert, and G. Carosi, Analysis of single-photon and linear amplifier detectors for microwave cavity dark matter axion searches, *Phys. Rev. D* **88**, 035020 (2013).
- [47] Z. Li, Squad_tuner data, Zenodo, 2025, [10.5281/zenodo.17210375](https://zenodo.org/record/17210375).